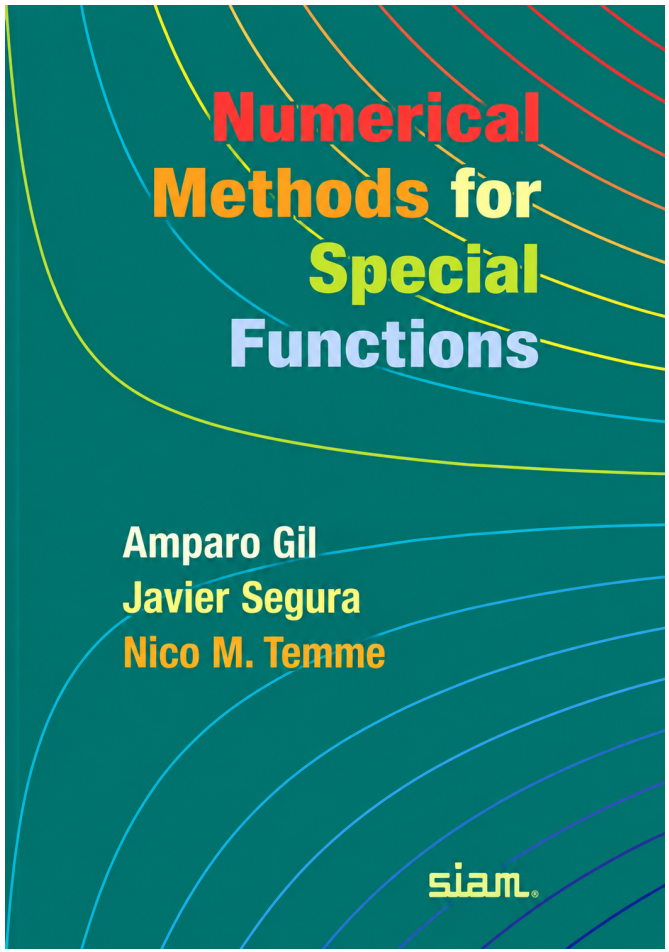
Spherical Harmonics and the Associated Legendre Functions of Integer and Half-Integer Degrees

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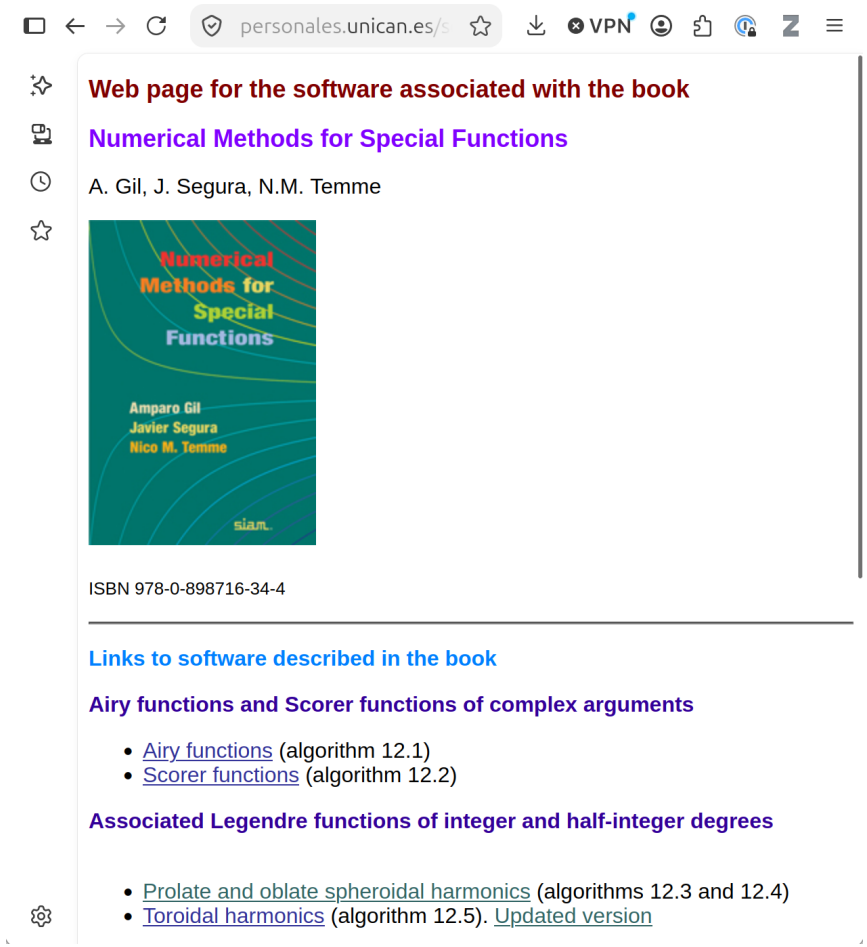
November 9, 2026

Numerical Methods for Special Functions



[4]

Author Website



Source Codes

Links to software described in the book

Airy and Scorer functions of complex arguments

- **Airy functions (Algorithm 12.1)**
- **Scorer functions (Algorithm 12.2)**

Associated Legendre functions of integer and half-integer degrees

- **Prolate and oblate spheroidal harmonics (Algorithms 12.3 and 12.4)**
- **Toroidal harmonics (Algorithm 12.5)**
  Updated version

Source Codes

Links to software described in the book

Modified Bessel functions

- Integer and half-integer orders (Algorithms 12.6 and 12.7)
- Purely imaginary order (Algorithm 12.8)

Parabolic cylinder functions

- Real order (Algorithm 12.9)

Zeros of Bessel functions

- Zeros of Bessel functions of the first kind (Algorithm 12.10) The book notes that fourth-order methods supersede this algorithm.

References II

[5] Nikolaos K. Pavlis et al. "The development and evaluation of the Earth Gravitational Model 2008 (EGM2008)". In: *J. Geophys. Res.* 117 (B4 Apr. 2012), 2011JB008916. ISSN: 0148-0227. DOI: 10.1029/2011JB008916. (Visited on 09/13/2026).

[6] J Segura and A Gil. "Evaluation of toroidal harmonics". In: *Computer Physics Communications* 124.1 (2000), pp. 104–122.

References II

[5] Nikolaos K. Pavlis et al. "The development and evaluation of the Earth Gravitational Model 2008 (EGM2008)". In: *J. Geophys. Res.* 117 (B4 Apr. 2012), 2011JB008916. ISSN: 0148-0227. DOI: 10.1029/2011JB008916. (Visited on 09/13/2026).

[6] J Segura and A Gil. "Evaluation of toroidal harmonics". In: *Computer Physics Communications* 124.1 (2000), pp. 104–122.